



EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

Ten-minute version

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The question

How many edges can a graph on n vertices carry
if *no* pair of vertices has m edge-disjoint paths?

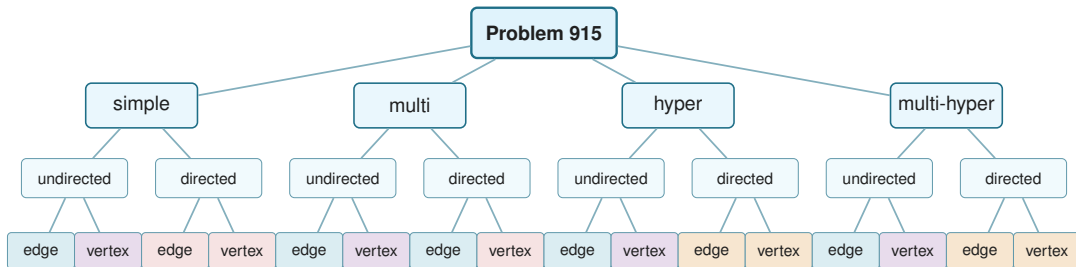
$$\ell_m(n) = \max \{ |E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1 \}$$

Mader, for simple graphs and all $n \geq 2, m \geq 2$:

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

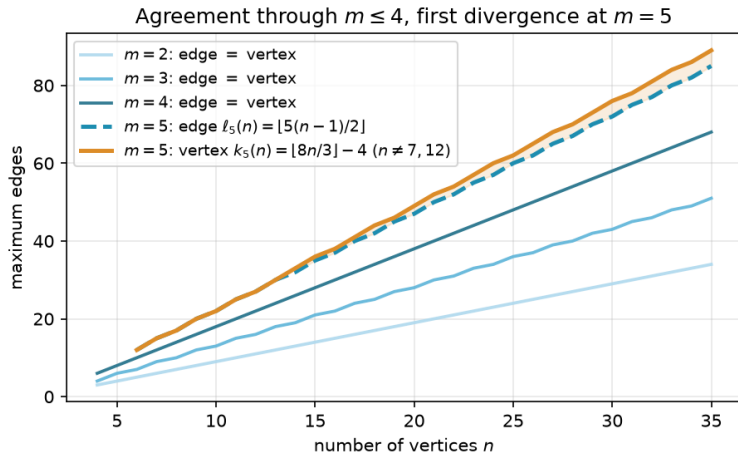
Proved here again from the Gomory–Hu tree, and a construction meets it for every $n \geq m$. Then three toggles on the model and one on the separation turn one question into sixteen.

Four models, two directions, two separations



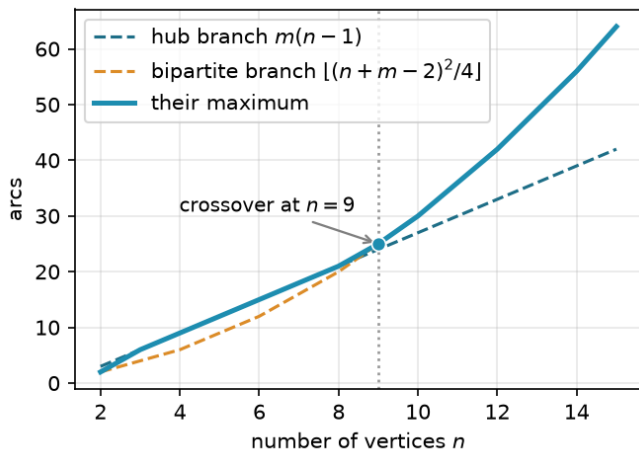
Arity, multiplicity, direction, separation. Sixteen leaves, and after this thesis they sit in four regimes rather than one.

Surprise one: edge and vertex routes part company



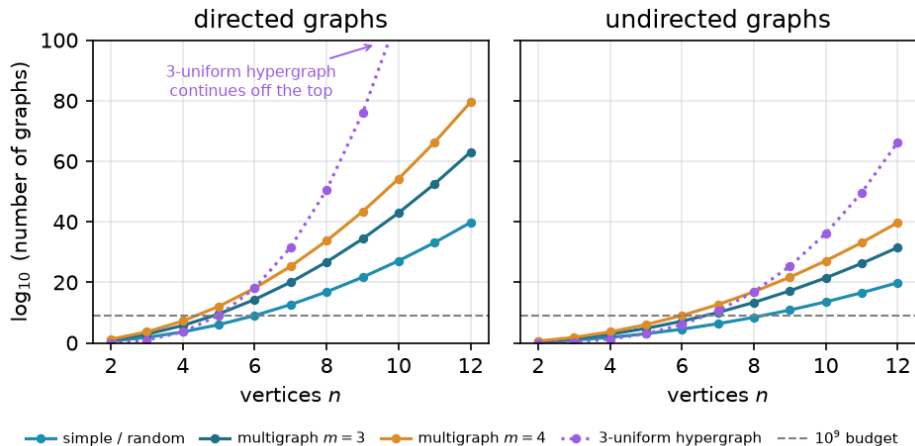
Identical through $m = 4$. At $m = 5$ the vertex value $\lfloor 8n/3 \rfloor - 4$ first passes the edge value at $n = 14$, and for $m \geq 6$ the vertex problem has stood open since 1974.

Surprise two: one-way traffic buys a quadratic



A hub carries $m(n-1)$ arcs, an augmented one-way bipartite wall carries $\lfloor (n+m-2)^2/4 \rfloor$. At $m=3$ the wall takes the lead near $n=9$, so the directed value is quadratic where the undirected one is linear.

Why a machine: enumeration detonates



Candidate counts on a logarithmic scale. Every model passes a generous 10^9 budget while n is still in single digits, and the directed cases, exactly the open ones, go first.

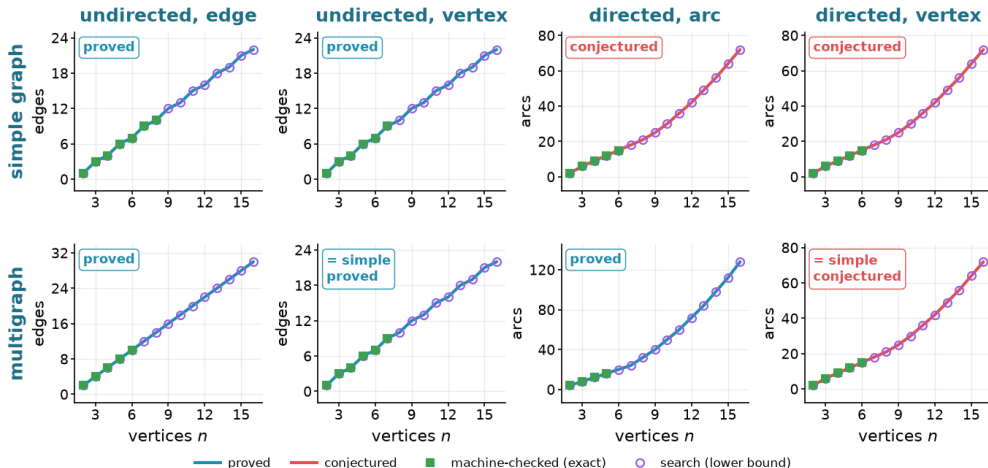
What the machine is allowed to claim

Construction	Variant	n	m	Search	Theory
spanning tree (simple undirected)	$\ell_m(n)$	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected)	$L_m(n)$	6	3	10	10
bidirected star (simple directed)	$\ell_m^{\text{dir}}(n)$	4	2	6	6
bidirected star (simple directed)	$\ell_m^{\text{dir}}(n)$	6	2	10	10
hub-bipartite tie (simple directed)	$\ell_m^{\text{dir}}(n)$	7	2	12	12
double star (multi directed)	$L_m^{\text{dir}}(n)$	4	3	12	12
double star (multi directed)	$L_m^{\text{dir}}(n)$	5	3	16	16

Started from the empty graph and told only to pack in edges, the tabu search returns the named constructions: the spanning tree, the thickened star, the bidirected star, the double star, and the hub-versus-wall tie at $n = 7$. It still only ever exhibits a graph, so it proposes a lower bound. A hand proof or a finished exhaustion disposes the upper bound, and only when the two meet is a value settled.

The eight graph-model variants at $m = 3$

Erdős 915 across the sixteen variants, $m = 3$ · simple and multigraph models



Blue a theorem, red a conjecture, yellow an interpolation. Filled squares are exact values, open circles are witnesses.

What is new here

1 One-way multigraphs, solved for every n and every m

$L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$, by deleting a sparse reachability-preserving subgraph once per unit of $m-1$ rather than a vertex at a time. The quadratic extremisers are classified for $n \geq \max(8, 2m)$.

2 One-way simple graphs: an unconditional quadratic bound

$|A(D)| \leq \lfloor \frac{n^2}{4} \rfloor + 4(m-1)(n-1)$, with no hypothesis on the shape of the extremiser, so the leading constant and the order of the next term are theorems. The same count re-runs under the stricter separation.

3 Hyperedge networks: the vertex problem at $m=2$ and $m=3$, and a directed leading term

$\lfloor 2(n-1)/(r-1) \rfloor$ is a universal $m=3$ bound from a new incidence-graph rank lemma, attained for $r \geq 3$ when $2 \leq \binom{n-2}{r-2}$. One-way, the leading term is $(m-1)n^2/(4(r-1))$ under both separations.

What is left

- The undirected *vertex* problem at $m \geq 6$, open since 1974
- The decomposition that would turn the directed conjecture into a theorem: that every extremal non-hub digraph splits as $A \cup B$ with no arc coming back
- Whether the directed vertex value equals the arc value at $m \geq 3$, which Whitney's inequality does not hand over
- The exact values behind the leading terms now settled, one-way and for hyperedges alike

The map is filled in where the arguments reached,
and the blanks are marked where they did not.